

More Symmetric Group Practice

1. Let $\alpha = (1\ 2\ 3)(4\ 5)(6\ 7\ 8\ 9)$ be an element of S_9 .

a) What is the order of α ? $|\alpha| = 12 = \text{LCM of lengths of disjoint cycles}$

$\alpha^{12} = \text{the identity permutation} = \text{LCM of } 3, 2 \text{ and } 4$

b) What is the order of α^2 ? Order of α^5 ? $|\alpha^5| = 12$

$|\alpha^2| = 6$ since $(\alpha^2)^6 = \alpha^{12} = \text{the identity}$

c) What is the connection between the powers of α and modular arithmetic?

$\alpha^i = \alpha^j$ iff $i \bmod n = j \bmod n$ where n is the order of α

d) What is α^{-1} ? For what positive power n is $\alpha^{-1} = \alpha^n$?

$\alpha^{-1} = (1\ 3\ 2)(4\ 5)(6\ 9\ 8\ 7)$. $\alpha \cdot \alpha^{-1} = e = \alpha^{12} = \alpha \cdot \alpha^{11}$ so by left cancellation of α we get $\alpha^{-1} = \alpha^{11}$

e) What is α^{100} ?

$$\alpha^{100} = \alpha^{12 \cdot 8 + 4} = (\alpha^{12})^8 \cdot \alpha^4 = e^8 \cdot \alpha^4 = \alpha^4$$

2. Write the following products of permutations in standard form and find the order of each.

a) $(1\ 2\ 3)(1\ 3\ 4)(1\ 4\ 5)(1\ 5\ 6) = (1)(2\ 3\ 4\ 5\ 6) = (2\ 3\ 4\ 5\ 6)$ has order 5

b) $(1\ 3\ 4)(1\ 5\ 6)(1\ 2\ 3)(1\ 4\ 5) = (1)(2\ 4\ 6\ 3\ 5) = (2\ 4\ 6\ 3\ 5)$ has order 5

b) $(1\ 2)(1\ 2\ 3)(1\ 2\ 3\ 4)(1\ 2\ 3\ 4\ 5)(1\ 2\ 3\ 4\ 5\ 6) = (1\ 4\ 3\ 5\ 6\ 2)$ has order 6

c) $(1\ 2\ 3\ 4\ 5\ 6)(1\ 2\ 3\ 4\ 5)(1\ 2\ 3\ 4)(1\ 2\ 3)(1\ 2) = (1\ 6)(2\ 5)(3\ 4)$ has order 2

3. Prove that S_n is not abelian if $n > 2$. (Hint: Find α and β such that $\alpha \cdot \beta \neq \beta \cdot \alpha$. Keep it simple!)

Let $\alpha = (1\ 2)$ and $\beta = (1\ 3)$. These are elements in S_n for all $n > 2$

$$\alpha\beta = (1\ 2)(1\ 3) = (1\ 3\ 2) \text{ but } \beta\alpha = (1\ 3)(1\ 2) = (1\ 2\ 3)$$

4. Find 8 permutations in S_4 that match up naturally with the elements of D_4 .

$$\begin{bmatrix} 2 & 1 \\ 3 & 4 \end{bmatrix}$$

D_4

R_0

R_{90}

R_{180}

R_{270}

H

V

D

D'

S_4

(1)

$(1\ 2\ 3\ 4)$

$(1\ 3)(2\ 4)$

$(1\ 4\ 3\ 2)$

$(1\ 4)(2\ 3)$

$(1\ 2)(3\ 4)$

$(1\ 3)$

$(2\ 4)$

Note: Other numberings for the square will lead to different permutations in S_4 .